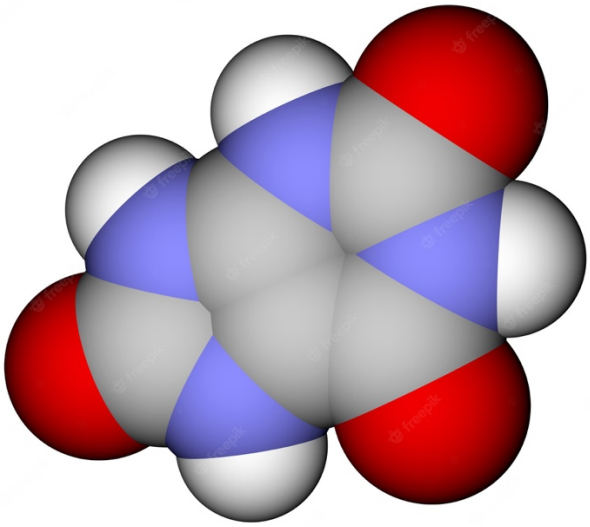
CHM 26300 – ORGANIC CHEMISTRY I



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Lab Time and Place

Course	Section	Time and Location
CHM 26300	411	R 7:30-10:20 am CHAS 4060
CHM 26300	412	R 7:30-10:20 am CHAS 4064
CHM 26300	421	R 11:30-2:20 pm CHAS 4060
CHM 26300	422	R 11:30-2:20 pm CHAS 4064
CHM 26300	431	R 3:00-5:50 pm CHAS 4060
CHM 26300	432	R 3:00-5:50 pm CHAS 4064
CHM 26300	433	R 3:00-5:50 pm CHAS 4069
CHM 26300	511	F 7:30-10:20 am CHAS 4060
CHM 26300	512	F 7:30-10:20 am CHAS 4064

Instructor

Dr. Mallory Kern CHAS 4074 mkkern@purdue.edu
Office Hour by appointment via email

Course Supervisor

Mr. Coriantumr (Cori) Wayment cwayment@purdue.edu

Graduate Instructors

TBA

Course Description

Laboratory experiments designed to illustrate the lecture material of CHM 26100. Elementary laboratory techniques essential to organic chemistry are introduced, followed by the exploration of organic compound separation, identification, synthesis and spectroscopy discussed in CHM 26100.

Learning Outcomes

1. Able to use theory to understand/predict experimental observations.
2. Able to demonstrate an understanding of the physical properties and a molecular understanding of chemical reactivity and materials.
3. Able to document scientific information and experimental data and write scientific reports, with graphical presentation of data.
4. Able to plan and perform chemical transformations.
5. Able to recognize and use chemical instrumentation and spectroscopy.
6. Able to identify and handle hazardous materials.

Required Text and Materials

J. W. Zubrick, The Organic Chem Lab Survival Manual, 10th edition
ISBN: 978-1118875780 E-book version: ISBN 978-1118875780
[Reserved in the Wilmeth Active Learning Center, 2nd Floor Library Desk].

You need goggles for the first lab sessions (see pages 11 and 12 to find what to wear in lab) and must have your own lock on your lab drawer by the 4th week of classes.

Digital Materials Charge: Students enrolled in this course must purchase digital materials for lab ($35). This digital content is required; students cannot complete the course without access to it. The materials will be released online on a real-time (approximately weekly) basis during the Fall 2025 semester.

You will purchase access to the digital materials via a Purdue University Online link
http://www.eventreg.purdue.edu/online/CHMFall35

Payment is due by September 8, 2025.

Other Reference Materials (in Library)

- Mayo, Pike, and Forbes: “Microscale Organic Laboratory Experiments”, 5th ed. John Wiley & Sons, Inc., 2011.
- L. M. Harwood, C. J. Moody, “Experimental Organic Chemistry”, Blackwell Scientific Publications, London, 1989.
- R. L. Shriner, R. C. Fuson, D. Y. Curtin, T. C. Morrill, “The Systematic Identification of Organic Compounds”, 6th Edition, John Wiley & Sons, Inc., New York, 1980.
- R. M. Silverstein, F. X. Webster, D.J. Kiemle “Spectrometric Identification of Organic Compounds”, 7th Edition, John Wiley & Sons, Inc., New York, 2005.
- J. Gordon, R. A. Ford, “The Chemist's Companion”, A Handbook of Practical Data, John Wiley & Sons, Inc., New York, 1973.

- CRC Handbook of Chemistry and Physics http://www.lib.purdue.edu/chem/resources/
- S. Budavari, Editor, The Merck Index, 12th Edition, Merck & Co., Inc., Whitehouse Station, New Jersey, 1996. http://www.lib.purdue.edu/chem/resources/
- Sigma Aldrich website. https://www.sigmaaldrich.com/united-states.html

WHAT TO EXPECT FOR LABORATORY SESSIONS

Prior to the laboratory session:

- You must carefully read the experimental description in Brightspace and the accompanying reading material cited at the beginning of the experiment. This material will be the basis for the Pre-Lab questions you must answer prior to coming to the lab.
- Each student must answer the **Pre-Lab** questions in Gradescope and submit to your TA no later than **1 hour** prior to the laboratory session. Failure to hand your Pre-Lab in on time will result in the loss of Pre-Lab points for that experiment (see grading section below).

At the beginning of the laboratory session:

- Pay full attention to the Pre-Lab introductory discussion given by your Graduate Instructor. S/he will describe the overall experiment, any procedural changes and highlight important issues related to safety and waste.

During the laboratory session:

- Complete the experiment (you will often work as a team of four with each team member conducting their own version of the experiment).
- Each student will record all requested data and observations within the **In-Lab** file of the experiment that you will download from Brightspace while you are performing the experiment. Obtain the data from the rest of your team when appropriate and enter this within the **In-Lab** pdf. Obtain data from the other teams within your lab room when appropriate and enter this within the **In-Lab** pdf.
- Each student will submit the updated version of the **In-Lab** file in Gradescope (containing all of your/team/lab room data) to your Graduate Instructor before you leave the laboratory. This is your **In-Lab** report.

After the laboratory session:

- Prepare your final report by filling in the Post-Lab section of the experiment in Gradescope. Note and submit this to your Graduate Instructor. **This is due by 11:59 pm a week after you completed the experiment.**

CAMPUS EMERGENCY INFORMATION

NOTE: Unanticipated circumstances may lead to changes in this schedule and the total points available for the course. In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be

necessitated by a revised semester calendar or other circumstances. In the event of emergency, contact Dr. Kern, the Course Supervisor or your Instructors about the necessary changes to the course that will be made. You are encouraged to familiarize yourself with Purdue Alert at:
www.purdue.edu/emergency_preparedness/

CHM 26300 Lab Experiments (Experiments 1 – 8)

Week	Experiments
1	Check in, Announcements
2	Deciphering Contributing Factors in the MeltingPoint Trends of Organic Solids – A Team Based Experiment **(Experiment 1)**
3	Deciphering Boiling Point Trends of Organic Liquids – A Team Based Experiment **(Experiment 2)**
4	Isolating Caffeine from Natural Sources (part 1) **(Experiment 3)**
5	Isolating Caffeine from Natural Sources (part 2) **(Experiment 3)**
6	Solvent and Polarity Effects in TLC **(Experiment 4)**
7	Separation of Stereoisomers Using TLC **(Experiment 5)**
8	No Experiments - Fall Break
9	Separation of Drug Components **(Experiment 6)**
10	Separation of Drug Components **(Experiment 6)**
11	Flash Chromatography Part 1 **(Experiment 7)**
12	Flash Chromatography Part 2 **(Experiment 7)**
13	Unknown Identification and NMR **(Experiment 8)**
14	No Experiments – Thanksgiving Break

15	Purification of Biphenyl (**Exp 9)**
16	Check-out
17	Finals Week (no lab final exam

Grading

- **Pre-Labs**. Pre-Labs are turned in **1 hour** prior to the start of the lab and are worth 20% of the total lab grade. If not submitted on time, the student is penalized **50% of the total** points of the Pre-Lab, but still must finish the Pre-Lab within 20 minutes of the start of lab to begin working in the lab. This policy is intended to ensure that students know what is going on and can work safely.
- **In-Labs** must be turned in before leaving the laboratory. Failure to turn in your In-Lab will result in the loss of **20%** of your grade per day for that particular experiment until no points remain.
- **Labs Reports** for Experiments 1-8 are each worth 50 points total (including Pre-Lab and In-Lab). Reports are due by 11:59 pm one week after the lab completion date.
- Reports not submitted by 9:00 pm are considered "late" and assessed a penalty of **20%** per day until no points remain.

- **Points** for experiments and total points for course:

Experiments 1 – 8	50 pts each	400 pts
Total		**400 pts**

- **Guaranteed Grades:**
  A+ >98% A 97 - 90% of cumulative score
  B+ >88% B 87 - 80% of cumulative score
  C+ >78% C 77 - 70% of cumulative score
  D+ >68% D 67 - 60% of cumulative score
  F < 59% of cumulative score

- All requests for **regrades** must be submitted to your TA ***within 2 weeks*** from the date the graded lab report was returned to the class. You must include a written explanation of the perceived problem together with the report. The whole report will be re-graded.

ABSENCE AND TARDINESS POLICY

NOTE!

It will not be possible to make up absences from a lab. NO EXCEPTIONS!!

A student can only miss TWO laboratory sessions during the semester without the approval of the course supervisor and Dr. Kern. If more than two sessions are missed (without approval), the student will automatically receive an F in the course.

Excused Absences

The following are the **ONLY** reasons for which an absence from a laboratory session *might* be excused if appropriate. Please let your lab supervisor know that you will be missing lab **prior** to the time of lab.

- **Illness (MEAPS). If you feel ill, do not come to lab, go to the Emergency Room and submit a doctor's note to the Office of the Dean of Students (ODOS)**
  - Please let your lab supervisor know that you will be missing lab **prior** to the time of lab.
  - You will be excused from that laboratory with your documented excuse and it will not negatively affect your grade.
- **A serious illness requiring medical attention.** A doctor's note must be provided to ODOS **no later than 1 week prior** to missing the lab experiment. After 1 week this will be considered an unexcused absence.
- **Death of an immediate family member.** The Course Supervisor must be able to verify the death and the student's attendance at the funeral services. **An absence form, and verifiable documentation, must be completed and submitted no later than ONE WEEK from the date of the absence.** Full details of procedures of the University Greif Absence Policy are available at: http://www.purdue.edu/odos/services/griefabsencepolicyforstudents.php
- **A direct conflict with another required* university activity.** An absence form for this type of conflict must be completed and submitted with an attached verification letter **at least one (1) week before** the conflict. Absences of this type will not be handled or considered after the conflict has occurred.

*Club activities will not be excused unless the activity is a professional activity directly associated with your major.

If you miss a lab due to documented illness as described above or COVID isolation/quarantine, **please note that you will still need to submit your PreLab report for the missing experiment on time, and obtain your InLab data from your

Tardy Policy

Day 1 of an Experiment

- A student who misses the Pre-Lab lecture will NOT be allowed to perform the experiment that day.
- No extra time will be allotted, and no experiments can be made up.

Day 2 of an Experiment

- The student who was not present on the first day of the experiment may participate in the later days ONLY after submitting her/his Pre-Lab write-up and consulting with a TA regarding safety concerns.
- The student must be no more than 10 minutes late to begin the experiment each day. No extra time will be allotted for tardy students.

Protect Purdue

The Protect Purdue Plan, which includes the Protect Purdue Pledge, is campus policy and as such all members of the Purdue community must comply with the required health and safety guidelines. Required behaviors in this class include: staying home and contacting the Protect Purdue Health Center (496-INFO) if you feel ill or know you have been exposed to the virus, avoiding shared use of personal items, maintaining robust hygiene (e.g., handwashing, disposal of tissues) prior to, during and after class, and following all safety directions from the instructor.

Students who are not engaging in these behaviors will be offered the opportunity to comply. If non-compliance continues, possible results include instructors asking the student to leave class and instructors dismissing the whole class. Students who do not comply with the required health behaviors are violating the University Code of Conduct and will be reported to the Dean of Students Office with sanctions ranging from educational requirements to dismissal from the university. Any student who has substantial reason to believe that another person in a campus room (e.g., classroom) is threatening the safety of others by not complying may leave the room without consequence. The student is encouraged to report the behavior to and discuss next steps with their instructor. Students also have the option of reporting the behavior to the Office of the Student Rights and Responsibilities. See also Purdue University Bill of Student Rights.

Academic Integrity

"As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern.

All students are expected to be familiar with Purdue's policies on academic integrity (https://www.purdue.edu/odos/academic-integrity/).
"Dishonesty in connection with any University activity may result in informal action or disciplinary sanctions. Cheating, plagiarism, or knowingly furnishing false information to

the University are examples of dishonesty. The commitment of acts of cheating, lying, stealing, and deceit in any of their diverse forms (such as the use of ghost-written papers, the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest." *From University Senate Document 72-18.*

Academic integrity means "doing your own work" at all times. Discussion of chemical concepts is encouraged, but sharing your answers and work on social media for the express purpose of letting other students copy it is not acceptable. Such a use of technology does not help you learn the material and is considered academic dishonesty. Using online resources such as Chegg to gain answers to lab questions is not allowed. Instructors can obtain user information from Chegg when lab questions are posted there.

Consequences of academic dishonesty include receiving a lower or failing grade for an assignment, receiving a lower or failing grade for the course and/or dismissal from the University. All incidents of academic integrity are referred to the Office of the Dean of Students. A student accused of academic dishonesty will be afforded due process as defined by Purdue University procedures.

Some examples of academic dishonesty are listed below. While this is not a complete list of academic dishonesty, these examples are provided for your information. If you have any questions about permissible behavior, you should ask before acting.

- Copying or possessing an unauthorized answer key.
- Copying from another student's work; allowing another student to copy your work.
- Copying lab data or a lab report; giving your data or lab report to someone else to copy (outside of the shared data exercises).
- Changing data for a lab to fit the perceived answer, that is, what you think the answer should be.
- Using someone else's data in a lab report as if it were your own (outside of the shared data exercises).
- Submitting a lab report or other work that you did not do.
- **The use of ChatGPT or other AI software to complete any portion of your pre- or post-lab assignments. THIS IS STRICTLY PROHIBITED. Dr. Kern WILL find out if you use it.**

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. More details are available on our course Brightspace table of contents, under University Policies.

This class will always be a place where individuals from every background, ethnicity, religion, country of origin, sexual preference, and gender identity will have a place to do their science without fear or harassment. Discrimination, racism, sexism, intolerance, and antagonism are never welcome here. We will always be in an inclusive environment, promoting empathy, diversity of people and thought. This class will only foster an atmosphere of positivity, encouragement, team work, and achievement. No matter who you are, you are welcome here.

Accessibility

The Disability Resource Center (DRC) is a resource for students and instructors. Students may present a "Letter of Accommodation" to you at any point in the semester. Should you have questions about accommodations, please contact the DRC at 765-494-1247 or email. Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let Dr. Kern know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Mental Health

Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 and http://www.purdue.edu/caps/ during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours.

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack, https://purdue.welltrack.com/. Sign in and find information and tools that are available to you at any time.

If you need support and information about options and resources, please see the Office of the Dean of Students, http://www.purdue.edu/odos, for walk-in hours (M – F, 8 am – 5 pm).

Safety in the Laboratory

See Brightspace and One Note for More "Detailed Safety Information"

Learn the location of ***Emergency Evacuation exits and routes*** and the ***emergency assembly area*** as posted in the laboratory. Be prepared and locate the nearest ***safety shower, eye wash fountain*** and ***fire extinguisher.***

Safety goggles (not safety glasses) are to be worn at all times in the laboratory, including check-out. Indirectly vented chemical splash goggles are required that meet the ANSI Z87.1:2020 (+D3) standard. No exposed holes on the sides, top of bottom that would allow chemicals or vapors to damage the eyes. Any student not wearing goggles will be required to leave, not be allowed to make up the experiment, and will receive a zero for the missed experiment.

It is a good idea to wear gloves at all times in the laboratory.

Students' ***clothing*** must cover their entire body from the neck to and including feet (*see picture on next page*). Tops/shirts must cover shoulders, underarms, and the entire abdominal area when standing, sitting or reaching. Pants/skirts must be long enough to reach the ankle when standing or sitting (tights or pantyhose cannot be used as a replacement). Sandals, shoes with open toe or heel, or openings in the clothing are not allowed. If a student comes to the laboratory inappropriately clothed, he/she must leave, obtain proper clothing, and return to complete the experiment. However, this can be done only once. For the second violation, the student cannot perform the experiment. ***Contact lens wearers*** are ***strongly*** encouraged to wear corrective glasses in the laboratory. Chemicals and vapors can lodge behind the contact lens and cause severe damage to the eye. Contact lenses also make the use of an eye wash ineffective. An exemption form must be completed and submitted to your Graduate Instructor before wearing contact lenses in the lab.

To avoid contact with reagents, flames and equipment, longer than shoulder length ***hair*** must be appropriately contained or restrained.

Personal electronic gear (including cell phones), food and beverages are not allowed!



WASTE DISPOSAL

Disposing of hazardous chemical waste has become an important issue in recent years. Proper disposal of hazardous waste is critical to the safety and well-being of our environment!

As a student in a chemistry lab course, you play an important part in this picture as YOU are the person generating and disposing of this hazardous chemical waste.

- Every time you create hazardous chemical waste, your lab manual will include information regarding the proper disposal of the waste. Be sure to follow these directions.
- **Read waste labels carefully!** Sometimes solids can be mixed with liquids, sometimes they cannot. The label will read "SOLID ONLY" or "LIQUID ONLY" if mixing of solids and liquids must not occur.
- If you are in doubt about the disposition of any waste, consult your Graduate Instructor first.
- If you can't find a particular waste container, look under the hood in the cabinets marked for your class, or ask your Graduate Instructor to help you.
- If the waste container you are about to use is full, ask your Graduate Instructor to get you a new one. **You must leave at least 4-6 cm of space from the top of the liquid to the top of the waste container.**
- **NEVER** pour anything into the sink unless you are directed to do so.
- When rinsing glassware with acetone and other solvents, use as little as possible. Pour the acetone/solvent rinses into the **appropriately labeled** large waste container located in the first main hood. **Do not pour organic solvents in the sink!**
- The acetone waste container is for acetone that contains *traces* of other solutes and solvents.
- **NEVER** put filter paper in any waste container unless you are directed to do so.
- **NEVER** throw anything in the trash can unless you are directed to do so.
- **IMMEDIATELY** clean up any spills that you make. Check with your Graduate Instructor regarding the proper procedures to use.

We must have your cooperation in these matters!